RAJALAKSHMI ENGINEERING COLLEGE

Keerthanaa S 250701346-2025-CSE

K2

Started on	Sunday, 30 August 2026, 4:29 PM
State	Finished
Completed on	Sunday, 30 August 2026, 4:35 PM
Time taken	6 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```
#include<stdio.h>
void quickSort(int arr[],int low,int high)
{
    if(low<high)
    {
        int i=low;
        int j=high;
        int pivot=arr[(low+high)/2];
        int temp;

        while(i<=j)
        {
            while(arr[i]<pivot)
            i++;
            while(arr[j]>pivot)
            j--;
            if(i<=j)
            {
                temp=arr[i];
                arr[i]=arr[j];
                arr[j]=temp;
                i++;
                j--;
            }

        }
        if(low<j)
        quickSort(arr,low,j);
        if(i<high)
        quickSort(arr,i,high);
    }
}
int main()
{
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    quickSort(arr,0,n-1);
    for(int i=0;i<n;i++)
    {
        printf("%d",arr[i]);
        if(i<n-1)
        printf(" ");
    }
    return 0;
}


```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course